Journal of Digital Intelligence and Economic Growth
Website: https://www.jdieg.com
Vol. 2 No. 4, December 2025, 89-116

ISSN (Print): 3058-3535
ISSN (Online): 3058-6518
DOI: https://doi.org/10.63768/jdieg.v2i4.006

GVC Integration, Local Value Chain Development and Manufacturing Innovation: The Dual Mechanisms of Human Capital and Finance

Guangyue Ge[1], Chen Lin [2], Zhuqing Mao [3 4] *

[1] (No. 42, Guijiang Lane, Suzhou Industrial Park, Suzhou) International Trade Project Manager, Harbin Electric Group Suzhou Dawson Valve Co., Ltd , Suzhou, China
[2] (NO.1900 Wenxiang Road, Songjiang District, Shanghai 201620) Master Student, School of International Business， Shanghai University of International Business and Economics, Shanghai, China
[3] (NO.1900 Wenxiang Road, Songjiang District, Shanghai 201620) Assistant Professor, School of International Business， Shanghai University of International Business and Economics, Shanghai, China
[4] (69 Kingsway Avenue JBS Park Auckland Park, Johannesburg 2092) Senior Research Associate, SARChI Industrial Development,， University of Johannesburg, Johannesburg, South Africa

Abstract： Amid rapid global economic development and deep restructuring of global value chains (GVCs), China's manufacturing sector confronts escalating challenges from rising labor costs, intensified international competition, and technological blockades imposed by developed countries in high-tech fields. This study investigates the driving effect of the domestic value-added ratio in exports (DVAR)，which indicates the development of local value chains, on manufacturing enterprises' innovation capabilities through the theoretical lens of regulatory mechanisms, specifically focusing on human resource supply optimization and financing environment improvement as critical pathways. While existing literature most focus on the effects of GVC participation on innovation or treats DVAR as a dependent variable, this research positions DVAR as an independent variable driving firm-level innovation. We hypothesize that local value chains enabled enterprises to accumulate greater innovation resources, with human resource supply and financing environment serving as crucial regulatory mechanisms. Employing fixed-effects models on Chinese manufacturing enterprise data, the empirical analysis demonstrates a significant positive correlation between local value chains and innovation capabilities,. The enhancement effect is particularly pronounced in technology-intensive enterprises. Robustness tests confirm strong consistency across alternative model specifications, while heterogeneity analysis reveals significant variations across different industry types. Theoretically, this study supplements existing GVC and innovation literature by establishing a regulatory mechanism framework linking DVAR to innovation capabilities. Practically, it provides a strategic pathway for China's manufacturing sector to address external challenges and achieve high-quality development through enhanced domestic value capture. The findings inform policymakers to strengthen support for innovation development strategies and promote deeper collaboration between education and capital markets to facilitate manufacturing transformation and upgrading. Future research should explore synergistic effects across diverse regional development contexts to achieve comprehensive improvement in enterprise innovation capabilities.

Key Words： GVC, Innovation Capability, Manufacturing Industry，Human Capital， Finance

*Fund: NO.2023ZJB008, Identification, Cultivation and Application of "Critical Points" in the Manufacturing Sector in the New Development Pattern, Shanghai Social Sciences Research Planning Project.
Author Correspondence: Guangyue Ge (geguangyue0814@163.com), Chen Lin (a15167726601@126.com), (Corresponding author) Zhuqing Mao(mm7ng7ng@hotmail.com)
Received 1 December 2025, Revised 20 December 2025, Accepted 28 December 2025

1 Introduction

1.1 Research Background

Economic globalization has driven the establishment and development of a global division of labor system based on global value chains. Following its reform and opening-up, China accelerated its industrialization process, gradually emerging as the world's largest manufacturing nation and integrating into the global division of labor. Leveraging low-cost advantages such as its “demographic dividend” and inexpensive land, China embedded itself in the global value chain through “low-end integration.” This approach fueled explosive growth in its merchandise trade volume, cementing its status as the “world's factory.”

However, China's manufacturing sector now faces dual pressures from both internal and external sources. Externally, the world is undergoing unprecedented transformation. With the advent of a new industrial revolution, advancements in information technology and the emergence of new energy sources are forcing a restructuring of manufacturing industries, driving a profound reconfiguration of global value chains. China confronts the dual challenges of the middle-income trap and the Thucydides Trap within this global value chain. On one hand, as per capita income rises, China must transition from labor-intensive industries to high-value-added capital- and technology-intensive sectors, facing fierce competition from developed economies in this process. On the other hand, Western nations are rolling out policies to seize commanding positions in the global value chain division of labor, imposing technological “chokepoints” on China. Simultaneously, some developing Southeast Asian countries are emerging, seeking to capture a share of low-end manufacturing. Internally, within the dual-circulation new development paradigm, the domestic circulation still faces numerous challenges, including slow industrial upgrading and regional development imbalances. These issues hinder the coordinated development of China's economy. Although China has achieved breakthroughs in some high-tech fields, overall industrial upgrading remains sluggish. Production models rooted in traditional manufacturing have yet to fully transform, and production efficiency urgently needs improvement. Secondly, regional development disparities remain pronounced. While eastern coastal regions experience rapid growth, central and western areas lag behind, constraining nationwide economic coordination.

To address these internal and external pressures, China must accelerate innovation capacity building, drive traditional industry transformation, and enhance the value-added of products and services. Only through such measures can Chinese manufacturing seize new opportunities in the global value chain restructuring and overcome current challenges.

Facing increasingly severe challenges, enhancing the innovation capacity of manufacturing has become crucial to breaking through the impasse. In 2006, the National Science and Technology Conference proposed a strategy of independent innovation and released the National Medium- and Long-Term Science and Technology Development Plan Outline (2006–2020). This aimed to create a favorable environment for independent innovation, promote enterprises as the main force in technological innovation, and commit to building an innovation-driven nation.

Since the implementation of the innovation-driven strategy, China's economic growth has gradually shifted from being “factor-driven” and “investment-driven” to “innovation-driven.” This study investigates how firms' capacity for domestic value creation within Global Value Chains (GVCs) drives innovation. Specifically, we focus on the Domestic Value Added Ratio (DVAR) in exports, the share of export value created domestically， as the core explanatory variable. DVAR reflects a firm's positioning and value capture capability within GVCs, not a separate "local value chain." The term "local value chain" in this paper is used metaphorically to emphasize the domestic linkages (supplier networks, knowledge spillovers, factor markets) that mediate the DVAR-innovation relationship. Our theoretical framework posits that enhancing domestic value creation in exports (higher DVAR) facilitates innovation through two domestic-channel mechanisms: human resource supply and financing environment optimization. This distinguishes our work from GVC embeddedness studies, shifting focus from global integration to domestic value capture as the innovation driver.

1.2 Research Significance

1.2.1 Theoretical Significance

This study focuses on the enterprise level, analyzing and summarizing the factors influencing innovation capabilities. It examines how the domestic value-added of enterprises within China's manufacturing industry supply chain integrates human resource supply and financing environment optimization to guide the direction of enterprise innovation capabilities, promote their development, and more rapidly and efficiently enhance the progress of innovation capabilities in China's manufacturing sector.

This study advances theoretical research on the relationship between local value chain and innovation capability, supplementing the link between the two. Existing literature predominantly treats the DVAR as an explanatory variable, exploring its influencing factors and determining mechanisms, while rarely examining the impact of DVAR on corporate behavior from the perspective of an explanatory variable. This study theoretically reveals the underlying mechanism by which the DVAR moderation effect drives corporate innovation capabilities. It not only shifts the research perspective on DVAR from a “result variable” to an “explanatory variable” but also uncovers its crucial relationship with corporate innovation capabilities. This paradigm shift provides a new theoretical framework for comprehensively understanding DVAR's role in corporate competitiveness and technological advancement.

Second, existing research has relatively neglected the role of local value chains in driving corporate innovation capabilities, focusing instead on the impact of global value chain embeddedness on innovation. This study constructs the impact mechanism of DVAR on innovation capability enhancement from the perspective of local value chains. By integrating DVAR with a moderation mechanism framework, the study further reveals the intrinsic logic of how local value chains drive industrial upgrading and technological progress through synergistic effects with human capital and finance, providing theoretical support for designing policies.

1.2.2 Practical Implications

First, this paper emphasizes that enhancing the domestic value-added ratio (DVAR) in exports of enterprises is a crucial pathway for China's manufacturing sector to gain competitive advantages amid the restructuring of global value chains. By optimizing human resource supply and financing environments, enterprises can strengthen technological innovation capabilities, elevate product value-added, and thereby secure a higher standing in international markets. This holds significant reference value for China's manufacturing sector in addressing international technological barriers and low-cost competition from Southeast Asia. Subsequent policy recommendations propose leveraging regulatory mechanisms to drive manufacturing upgrading. This offers direction for national and local governments: for instance, encouraging enterprises to increase DVAR through fiscal subsidies, R&D support, and tax incentives, with a focus on technology-intensive industries; and establishing robust financial support systems to address SMEs' financing challenges. The theoretical framework and empirical findings presented in this study clarify the mechanisms through which optimizing local value chains enhances manufacturing innovation capabilities. This provides practical guidance for China's transition from traditional manufacturing models to high-quality development, particularly by building an industrial ecosystem that jointly develops human resources and financing environments to elevate overall manufacturing productivity and technological standards.

Addressing China's manufacturing sector challenges of rising labor costs and international technology barriers, the findings emphasize that enhancing DVAR effectively accumulates innovation resources and drives corporate technological advancement, offering a crucial approach to resolving critical bottlenecks. As the global economy accelerates and global value chains undergo restructuring, China's manufacturing role within the global economic system has undergone profound transformation. As the global manufacturing leader, China has undergone a transformation from low-end to high-end manufacturing. However, rising labor costs, shifting international market demands, and technological barriers have presented unprecedented challenges and pressures. Addressing these challenges by achieving industrial upgrading and enhancing innovation capabilities has become a critical imperative for China's manufacturing sector.

1.3 Current Situation Analysis

1.3.1 Global Economic Trends and Changes in the Global Value Chain

Since the 1980s, globalization has brought significant transformations to the world economic landscape. Driven by technological revolutions and the advancement of information technology, the Global Value Chain (GVC) has undergone profound changes. Manufacturing production is no longer confined to a single nation or region but is distributed across every corner of the globe. Through its strategy of "integrating into global value chains," China rapidly developed into the world's factory, leveraging low-cost labor and resource advantages to achieve swift economic ascent. However, as GVCs undergo profound restructuring and production factors circulate globally, traditional low-cost advantages are gradually eroding. This forces China's manufacturing sector to confront the challenges of industrial upgrading and innovation.

The restructuring of global value chains has intensified competition among nations, particularly in technology-intensive and innovation-driven industries, which have become the primary battlegrounds. Simultaneously, technological barriers imposed by Western nations on China's high-tech industries, coupled with the rise of emerging economies in low-end manufacturing, have placed China's global market position under severe pressure. Consequently, enhancing local value chains, accelerating industrial transformation, achieving technological innovation, and diversifying markets have become critical issues in China's manufacturing development that cannot be overlooked.

1.3.2 Current Status and Challenges of China's Manufacturing Sector

China's manufacturing sector has achieved remarkable development in recent years but also faces multiple challenges. From a macro perspective, the overall development of China's manufacturing remains in a transitional phase, with low-end manufacturing still dominating. Although some high-end manufacturing sectors have made breakthroughs in certain areas, the industrial chain has yet to fully upgrade toward technology-intensive industries. Innovation capabilities remain relatively weak, and the sector continues to rely on traditional production models and cost advantages.

First, rising labor costs represent a major challenge for China's manufacturing sector. The production model historically reliant on low-wage labor can no longer sustain its competitive edge globally. As China's economy develops, workers' income levels have steadily increased, leaving enterprises grappling with recruitment difficulties and wage inflation. Many labor-intensive industries, particularly textiles and furniture manufacturing, are experiencing declining competitiveness, with a growing trend toward relocation to lower-cost regions like Southeast Asia.

Second, heightened uncertainty in the global trade environment—particularly escalating Sino-US trade friction and increasing trade barriers—has exposed Chinese manufacturing exports to greater market risks. As nations worldwide raise technological barriers and restrict exports of critical technologies and core components, China's manufacturing sector faces increasingly prominent technological constraints. This gap is especially pronounced in high-end technology fields like semiconductors, communications, and aerospace, where China's technological capabilities still lag significantly behind global leaders.

Moreover, the complexity of both domestic and international economic environments compels China's manufacturing sector to navigate global competition while simultaneously addressing the challenges of domestic industrial upgrading. China's traditional manufacturing is largely concentrated in low-value-added production, struggling to support the R&D and manufacturing of high-tech, high-value-added products. An irrational industrial structure, an imperfect technological innovation system, and relatively weak basic research have resulted in a severe shortage of innovation capabilities within the manufacturing sector.

1.3.3 Impact of Local Value Chains on China's Manufacturing Innovation Capacity

With the deepening integration and division of labor within global value chains, China not only faces external challenges but also plays a crucial role in the global industrial chain. Over the past few decades, operating as the "world's factory," China has exported vast quantities of low-value-added products globally at low prices. However, this model has gradually revealed its limitations in the context of globalization. Particularly as China confronts increasingly fierce competition in global markets, it must respond by enhancing technological innovation and increasing product value-added.

The Domestic Value Added Ratio in Exports (DVAR) serves as a crucial metric for assessing a nation's competitiveness

within global value chains. It reflects not only a country's production efficiency and technological capabilities but also its strategic position within these chains. China's manufacturing sector, historically reliant on low-value-added activities , finds it increasingly difficult to sustain its competitive edge. To secure a more advantageous position in global markets, China must intensify technological innovation and elevate product value-added.

By elevating domestic value-added, China's manufacturing sector can not only strengthen its competitiveness within global industrial chains but also drive technological advancement and innovation among enterprises. Research indicates that increasing the domestic value-added rate stimulates corporate technological innovation and R&D capabilities. This shift enables China's manufacturing industry to move beyond reliance on traditional low-cost production, instead achieving greater market share globally through independent innovation and technological upgrading.

1.3.4 The Relationship Between Regional Development Imbalances and Innovation Capacity

Regional imbalances in China's manufacturing sector remain pronounced. While manufacturing in eastern coastal regions is relatively advanced, central and western regions lag significantly in industrial development, resulting in marked disparities in national innovation capacity. China must implement policy measures to enhance innovation capabilities in central and western manufacturing sectors. Strengthening infrastructure, optimizing the business environment, facilitating technology transfer, and promoting industrial collaboration can help these regions elevate their innovation capacity and narrow the gap with eastern areas.

2 Literature Review

2.1 Research on Innovation Capability

The concept of “innovation” was first introduced by economist Joseph Schumpeter in his 1912 publication The Theory of Economic Development. He defined innovation as the creation of a new production function, primarily manifested through novel combinations of elements such as new products, new markets, new methods, new materials and their sources, and organizational structures. Over time, the definition of innovation has expanded, and “innovation capability” has emerged as a significant topic in academic research.

Innovation capability is generally understood as a firm's ability to identify and acquire external new knowledge, or to generate endogenous new knowledge with market value by recombining or applying existing knowledge. Achieving this process requires a series of strategies, organizational structures, technologies, and market practices (Chen Litian et al., 2012). This paper focuses on exploring the mechanisms for cultivating innovation capability.

Academic research typically categorizes the influencing factors of innovation capability into three levels: national, industrial, and corporate.

At the national level, Xie Yumin et al. (2014) conducted a more detailed analysis of previous studies on the reverse spillover effects of OFDI. Empirical results indicate that China's outward foreign direct investment (OFDI) only generates reverse spillover effects on domestic imitative innovation capability, while inhibiting autonomous innovation and secondary innovation capabilities, thereby suppressing the nation's overall innovation capability. Zhang Yang et al. (2015) demonstrate that factors such as national intellectual property protection, patent stock, R&D subsidies, and information technology capital significantly enhance national innovation capacity. Overreliance on external resources or indiscriminate foreign investment attraction not only fails to boost innovation capabilities but may even generate distortionary effects. Additionally, public perceptions of technology's importance and impact on quality of life positively influence national innovation capacity. Wu Dan et al. (2018), through comparative analysis between China and ten other countries, identified factors such as R&D intensity, number of R&D personnel, patent applications, scientific journal publications, and high-tech product exports as significantly influencing national innovation capacity.

At the industrial level, Wu Baiyu et al. (2016) categorized manufacturing sectors based on distinct characteristics into labor-intensive, capital-intensive, and technology-intensive industries. They introduced the concept of “other technological

activities," highlighting that R&D investment, foreign capital investment, and technological activities exert varying impacts on different manufacturing types. Liu Zhangsheng et al. (2017) demonstrate that government support, foreign investment, and enterprise scale positively correlate with China's manufacturing sector's green technological innovation capacity, while the business environment exerts a negative influence. Qiu Shilei et al. (2018) proposed that linkage effects, demonstration effects, competition effects, and personnel mobility effects influence innovation capabilities. They argued that linkage effects positively impact technological innovation through exports from high-tech industries, while demonstration effects have a relatively weaker impact. Competition effects promote technological innovation via R&D investment by foreign-funded enterprises, and personnel mobility effects indicate that highly skilled talent primarily flows from foreign-funded enterprises to domestic enterprises. Xie Cong et al. (2022) examined the spatiotemporal patterns, suggesting that factors such as economic development levels, environmental regulations, human capital, education levels, and resource endowments exert varying impacts on industrial innovation capabilities across different spatiotemporal contexts.

At the firm level, Zhang Jie et al. (2007) found through a large-scale survey that firm-scale expansion, measured by fixed assets and sales revenue, contributes to enhancing Chinese firms' independent innovation capabilities and innovation activities. Contrary to conventional wisdom, they also observed that agglomeration effects exert bidirectional influence on corporate innovation intensity, while export factors exhibit an inverted U-shaped nonlinear relationship with innovation intensity. Sha Wenbing et al. (2011) contend that domestic enterprises' independent R&D investment is pivotal for enhancing innovation capabilities, whereas foreign enterprises' R&D activities drive innovation through knowledge spillover effects—a phenomenon closely linked to industry openness to foreign investment. Dong Youde et al. (2014) proposed that OFDI significantly enhances Chinese enterprises' innovation capabilities, particularly through reverse spillover effects in manufacturing, operations, and R&D segments. Conversely, goods import-export trade inhibits corporate innovation capabilities, a conclusion unaffected by enterprises' OFDI behavior. Studies by He Qiong et al. (2022) and Li Jian et al. (2022) demonstrate that the development of the digital economy significantly enhances corporate innovation capabilities. Zhao Shengmin et al. (2023), approaching the issue from the perspective of corporate social responsibility, argue that social responsibility exerts a significant positive impact on corporate innovation investment, output, and efficiency by alleviating financing constraints, reducing financial risks, and mitigating agency problems.

2.2 Research on the Domestic Value-Added Rate of Corporate Exports

Domestic value-added in exports refers to the value entirely produced and contributed domestically within a country or region's exported products. Based on this concept, the domestic value-added rate of exports can be understood as the proportion of a country's contribution to the total value of its exported goods (Li Nan, 2022).

Against the backdrop of deepening international division of labor, a single country often cannot independently complete the production of a product. Instead, it relies on division of labor and collaboration among different countries to achieve this. Consequently, final product transactions between countries have gradually shifted toward the exchange of intermediate goods. This shift raises the question of how to accurately measure a country's true output value. In response, the export domestic value-added rate has emerged as a crucial indicator for assessing this value contribution.

Currently, many domestic scholars have shifted their research focus to the micro level, delving into the factors influencing the domestic value-added rate of corporate exports. Lv Yue et al. (2018) argue that market segmentation exerts a significant negative impact on the domestic value-added rate of corporate exports. In contrast, Gao Xiang et al. (2018) contend that distortions in factor markets substantially increase the domestic value-added rate of Chinese corporate exports. Their mechanism posits that the "relative price effect" arising from factor market distortions outweighs the negative "cost-plus effect" on the domestic value-added rate of corporate exports. Wang Yi et al. (2019) propose that environmental regulations significantly enhance the domestic value-added rate of enterprises' exports, with this effect being more pronounced for non-state-owned enterprises. Liu Yuhai et al. (2020) find that the VAT transformation policy significantly increases the domestic value-added rate of enterprises' exports through two channels: alleviating financing constraints and improving the capital-labor

ratio. This effect exhibits a dynamic pattern of rising initially and then declining. Shao Chaodui et al. (2020) indicate that the policy opening foreign equity participation in the service sector significantly enhances firms' export domestic value-added rate by lowering the relative price of domestic intermediate goods and increasing firms' cost-plus rate, while a favorable regional institutional environment further amplifies this effect; Sheng Bin and Wang Hao (2022), examining the geographic structure of financial supply, argue that the expansion of bank branches significantly boosts the domestic value-added rate of exports through the transmission mechanisms of "bank competition" and "bank-firm distance." Liu Huizheng et al. (2022) contend that trade digitization promotes the growth of the domestic value-added rate of exports through cost-addition effects and relative intermediate goods price effects. Wu Chuhao and Tang Jing (2024) find a significant negative correlation between the domestic-export value-added ratio of Chinese enterprises and the gap in digital technology between China and other countries, while identifying a significant positive relationship between technological progress and this ratio.

2.3 Research on the Relationship Between Innovation and the Domestic Value-Added Ratio of Enterprise Exports

Within the context of global value chain (GVC) division of labor, the domestic value-added ratio of exports reflects an enterprise's ability to capture value within international division of labor. In recent years, Chinese scholars have conducted in-depth research on how innovation influences DVAR.

Domestic scholars have produced extensive studies on innovation's impact on DVAR: Li Shengqi and Mao Qilin (2017) found that corporate R&D innovation positively correlates with DVAR.

Innovation enhances the variety and quality of domestic intermediate goods while lowering their relative prices, thereby encouraging firms to use more domestic intermediate goods instead of imported ones, ultimately boosting export domestic value-added rate. Conversely, upstream monopolies suppress downstream firms' R&D investment, lowering their cost markup rates and thereby reducing the proportion of domestic intermediate goods used, which decreases DVAR. Jiao Yanan (2019) argues that corporate innovation enhances China's labor productivity by improving worker quality, reducing labor intensity, and strengthening managerial capabilities. Furthermore, innovation boosts capital productivity by upgrading production machinery and improving capital allocation efficiency. Finally, corporate innovation enhances raw material productivity through resource reduction and recycling. Collectively, these factors optimize resource allocation, thereby boosting DVAR.

Zhang Yingying et al. (2020) found that technological innovation lowers firms' marginal production costs while enabling higher product pricing, increasing cost markup rates and directly elevating DVAR. Zhao Jingrui et al. (2021) argue that technological advances in upstream firms, through intermediate goods technology spillovers, prompt downstream firms to reduce reliance on imported intermediates and increase domestic intermediate usage, thereby raising DVAR.

Regarding markup rates and pricing power, Zhao Jingrui et al. share similar perspectives with Zhang Yingying et al. Gao Qiang and Song Lin (2022) propose that innovation, by expanding the variety of intermediate goods, exerts three distinct effects on DVAR.First, innovation broadens the production scope of domestic intermediate goods, intensifies domestic market competition, lowers domestic intermediate prices, and encourages enterprises to purchase more domestic intermediates, thereby increasing DVAR. Second, while innovation expands the variety of imported intermediates, heightened price competition among imports leads enterprises to rely more heavily on imports, consequently reducing DVAR.

Finally, innovation significantly expands the variety of domestic intermediates far more than imported ones, substantially increasing DVAR through price and substitution elasticity channels. Lv Yue et al. (2022), focusing on the service sector, categorize innovation into low-tech and high-tech innovation, finding low-tech innovation has a more pronounced effect on boosting DVAR. Such innovations have lower barriers to entry, making it easier for firms to achieve technological improvements through cost savings and technology spillovers brought by service sector opening. High-tech innovation (patent-protected inventions), however, exhibits weaker dynamic adjustment effects from service sector opening due to higher R&D complexity and capital requirements, resulting in insignificant direct impacts on DVAR. Song Lin et al. (2025) reveal multidimensional effects of digital technology innovation on manufacturing firms' domestic value-added ratio in exports (DVAR). First, digital technology innovation directly stimulates DVAR growth. Second, it indirectly enhances DVAR growth

through three mechanisms: boosting total factor productivity and reducing costs, decreasing reliance on imported intermediate goods, and driving product diversification and quality upgrades. Heterogeneity studies indicate this mechanism is more pronounced in SMEs, where innovation breakthroughs significantly impact competitiveness.

Foreign scholars have also produced substantial research focusing on this relationship, with many studies challenging conventional wisdom by examining the impact of DVA (FVA) on innovation. Keun Lee et al. (2017) classify a country's development stage, noting that as it reaches the high-income stage, foreign value added increases as innovation capabilities enhance reintegration into global value chains. The mechanism involves enterprises establishing local value chains and then achieving reintegration into global value chains through higher levels of openness. Regression results show a significant negative correlation between the extent of local knowledge creation and diffusion and foreign value added (FVA). The intermediary chain through which local knowledge promotes economic growth manifests as the transmission process: "local knowledge accumulation → enhanced local value creation capacity → accelerated economic growth." This paper's conclusions are highly representative. Regarding domestic knowledge creation, Haakonsson and Slepniov (2018) complemented the interaction between GVCs and innovation systems, demonstrating that robust local innovation systems facilitate the absorption and transformation of external technologies. Meanwhile, Keijser and Iizuka (2017) expand pathways for GVC learning from regional and local value chain perspectives, revealing the role of local and regional value chains in technological capability building. Regarding the relationship between FVA and innovation, Lema et al. (2018) propose a coevolutionary model that provides a systematic theoretical framework, arguing that the interaction between GVCs and innovation systems is not static but rather a dynamic, time-varying process. This model deepens our understanding of the relationship between GVCs and innovation systems, demonstrating that co-evolution can drive technological upgrading at both the national and firm levels.

2.4 Literature Review

Current domestic research on factors influencing corporate innovation capabilities, domestic value-added rates of exports, and their interrelationship provides crucial reference points for this study. While research on innovation capability determinants is extensive, few studies examine how talent and financing mechanisms moderate the impact of value chains on innovation capabilities.

Second, most domestic studies treat the domestic value-added ratio in exports (DVAR) as the dependent variable, with mechanisms often involving direct effects, intermediate goods substitution effects, and productivity enhancements. These studies primarily focus on innovation's role in driving local industrial development—such as how technological progress increases product value-added, optimizes industrial structure, or strengthens corporate global competitiveness. Few studies explore the potential role of local industrial development in driving innovation in reverse.

In fact, existing international research has examined the impact of DVAR on innovation, focusing on African and East Asian countries (Japan and South Korea). It is important to recognize that industrial development, by increasing the domestic value-added rate of exports, not only enhances resource allocation efficiency but also enables enterprises to accumulate more capital, technology, and talent required for innovation, thereby providing crucial support for innovative activities. Deepening our understanding of this bidirectional relationship and its underlying mechanisms will help comprehensively grasp the virtuous cycle between industrial development and corporate innovation, providing theoretical foundations and practical pathways for promoting high-quality development.

It is crucial to recognize that increasing the domestic value-added rate of corporate exports is not an end in itself. Rather, how can this improvement positively impact China's production and address the challenges currently faced by the nation?

Therefore, building upon prior research, this paper examines the pathways through which the domestic value-added rate of corporate exports influences the innovation capabilities of manufacturing enterprises, focusing on the regulatory mechanisms of human resource supply and financing environment optimization.

3 Theoretical Foundations of Innovation Capability

3.1 Definition of Innovation Capability

Innovation capability is typically defined as an enterprise's capacity to enhance its competitiveness and market adaptability through continuous technological innovation, management innovation, service innovation, and other means within an intensely competitive market environment. This capability encompasses both improvements to existing products, processes, and market models, as well as the creation of high-value-added innovations through the development of new technologies, products, and management concepts. In the manufacturing sector, enhancing innovation capability directly determines whether enterprises can leap from the “low end” to the “high end” of the global value chain and effectively drive industrial upgrading. Innovation capability extends beyond technological R&D and product innovation to encompass innovations in production management, market operations, and service systems. The innovation capabilities of manufacturing enterprises typically manifest as follows:

Technological Innovation Capability: Refers to an enterprise's ability to enhance product value-added and production efficiency by developing new technologies and products and improving production processes. The level of technological innovation capability directly impacts an enterprise's competitiveness in the market. Management Innovation Capability: Includes innovation in management processes, corporate organizational structures, and business models. Management innovation can help enterprises improve resource allocation efficiency, reduce costs, and enhance their ability to respond to market changes.

Market Innovation Capability: Market innovation involves developing new market opportunities through product customization, brand building, and other strategies based on market demand insights, thereby increasing market share.

3.2 Schumpeter’s Innovation Theory

Joseph Schumpeter’s innovation theory is widely regarded as the foundational work of innovation economics. In The Theory of Economic Development, Schumpeter posited that innovation is the core driver of economic progress. He emphasized that innovation extends beyond mere product or technological refinement, serving as the fundamental force propelling dynamic change within economic systems. He defined innovation as “new combinations”—the creation of market-competitive new products or services through novel combinations of production factors and the adoption of new production methods.

Schumpeter’s innovation theory specifically encompasses five types of innovation:

1. Development of new products: Enterprises introduce entirely new products or enhance existing ones through technological R&D and innovation to address unmet market needs.

2. Application of new production processes: Enterprises adopt novel production methods or techniques to boost efficiency and reduce costs.

3. Opening of new markets: Enterprises expand market share by entering new markets or establishing new distribution channels.

4. Use of new raw materials: Enterprises substitute traditional resources with novel materials or resources to enhance product innovation and competitiveness.

5. Innovation in organizational structures: Enterprises innovate in management and organizational frameworks to improve market adaptability through more efficient resource allocation and decision-making mechanisms.

Schumpeter posited that innovation drives economic development, yet innovation is not incremental but propels continuous economic evolution through “creative destruction.” In manufacturing, enterprises enhance productivity, increase value-added, and boost global competitiveness through technological innovation, process improvements, and market expansion. Schumpeter's innovation theory provides the theoretical foundation for your research in analyzing how enterprises enhance competitiveness by increasing their domestic value-added ratio in exports (DVAR) and innovation capabilities.

3.3 Innovation Theory in Endogenous Growth Theory

Endogenous Growth Theory, proposed by scholars such as Paul Romer and Robert Lucas, represents an extension of traditional economic growth theory. Unlike exogenous growth theories represented by the Solow model, endogenous growth theory emphasizes that the driving forces behind technological progress and economic growth originate from factors within the

economic system itself, rather than relying on exogenous variables. This theory incorporates endogenous factors such as innovation, knowledge accumulation, and human capital into growth models, revealing the intrinsic mechanisms of long-term economic growth and laying a crucial foundation for modern economic growth theory.

The core tenets of endogenous growth theory revolve around the roles of technological progress, knowledge accumulation, and human capital. Traditional growth models treat technological progress as an exogenous variable, assuming it cannot be directly influenced by economic policies or market actions. In contrast, endogenous growth theory internalizes technological progress, positing that firms and individuals can drive technological advancement through activities like R&D, innovation, and education, thereby fostering sustained economic growth. In Romer's growth model, knowledge is treated as a non-rival good with significant externalities and spillover effects. This implies that knowledge accumulated by firms in pursuit of profit maximization not only benefits their own development but also enhances overall economic productivity through spillover effects. Lucas further emphasized the role of human capital in economic growth, arguing that education and skill accumulation boost labor productivity, providing the driving force for long-term economic expansion.

The key mechanisms of endogenous growth theory are as follows:

1. Endogenous Technological Innovation

The endogenous growth theory emphasizes that technological innovation does not originate from external conditions but is the result of the behavior of firms and individuals within the economic system. Through R&D investment and innovation activities, firms can continuously improve production efficiency, reduce production costs, and create new products and services. For example, in knowledge-intensive industries, firms' R&D activities not only drive their own technological progress but also enhance the overall productivity of the industry and region through technology diffusion effects.

2. Knowledge Accumulation and Spillover Effects

Knowledge is a non-rivalrous and partially shareable resource whose accumulation generates significant positive externalities. A firm's technological innovations and knowledge accumulation not only enhance its own competitiveness but also benefit other firms and industries through imitation, learning, and technology transfer. These spillover effects make knowledge accumulation a vital driver of overall economic growth.

3. Enhancement of Human Capital

The accumulation of human capital is a vital component of endogenous growth theory. Through education, training, and skill enhancement, the quality of the labor force improves, leading to increased innovation capacity and productivity. Education not only elevates individual productivity but also boosts the technological level and economic vitality of society as a whole through knowledge dissemination and skill sharing.

4. Increasing Returns to Investment Scale

Endogenous growth theory posits that returns on investment in knowledge and human capital may exhibit increasing returns. The diminishing returns principle in traditional economics is modified within this framework, as knowledge and technological innovation possess accumulative effects rather than physical limitations inherent to conventional capital. This increasing-returns dynamic enables sustained growth within economic systems.

Theoretically, endogenous growth theory expands our understanding of economic growth mechanisms in several ways. First, it transforms technological innovation and knowledge accumulation from exogenous variables into endogenous factors, revealing the intrinsic drivers of economic growth. This shift enables policymakers to directly influence growth through R&D support, educational investment, and institutional innovation.

Second, endogenous growth theory emphasizes the spillover effects of knowledge and technology. This perspective not only explains the sustained growth of technologically advanced nations and regions but also provides a theoretical basis for developing countries to achieve catch-up through technology imitation and introduction.

Finally, the theory elevates the importance of human capital to a new level, highlighting that the accumulation of education and skills is not only crucial for individual development but also the foundation for a nation's sustained economic growth.

4 Mechanisms and Hypotheses Proposed

4.1 Proposed Mechanisms

In recent years, the acceleration of economic globalization and industrial upgrading has driven the shift of manufacturing toward higher-value segments of the global value chain. Against this backdrop, China's manufacturing sector has gradually developed a multi-tiered innovation capability enhancement mechanism driven by local value chains.

First, value chains provide crucial resource support for corporate innovation. Through close collaboration with upstream and downstream partners in the industrial chain, manufacturing enterprises achieve effective knowledge and technology transfer and transformation, enabling smooth innovation activities.

upstream and downstream partners, enabling effective knowledge and technology transfer and transformation, thereby facilitating innovation activities. Amid intensifying competition in the global value chain, enterprises enhance the domestic value-added ratio (DVAR) in export products. Through industrial chain integration, they not only strengthen technological accumulation but also promote coordinated domestic industrial development, laying a solid foundation for innovation capability enhancement.

Second, human resource supply plays a pivotal role in driving innovation. The innovation capacity of domestic manufacturing enterprises heavily relies on the availability of highly skilled talent. Continuous improvement in human resources enhances R&D efficiency by elevating the human capital required for innovation. Simultaneously, the interaction between human resource supply and innovation outcomes can promote the localized application of technology, enabling enterprises to better meet the skill demands of high-value-added production. Therefore, the rise in domestic value-added rates creates favorable conditions for enhancing innovation capabilities through the regulatory mechanisms of industrial chains and human resource supply.

Finally, the optimization of the financing environment ensures the financial security of innovation activities. Particularly for high-investment innovation projects, an improved financing environment enhances enterprises' access to capital and financial stability. When manufacturing enterprises increase the domestic value-added rate of export products, they typically face substantial capital requirements for technological R&D and innovation.

A stable financing environment enables enterprises to mitigate capital turnover risks and sustain innovation activities.

Thus, the local value chain forms a systematic innovation driver through dual regulatory mechanisms: human resource supply and financing environment optimization.

4.2 Research Hypotheses

Based on the theoretical mechanism of local value chain-driven innovation capability and existing literature, this study proposes the following hypotheses:

Hypothesis 1: A positive correlation exists between a firm's export domestic value-added rate and its innovation capability. As the domestic value-added rate increases, the firm's value capture capacity improves. Concurrently, knowledge spillovers from industrial agglomeration and optimized industrial structures attract greater resource allocation, thereby enhancing innovation capability. The rise in value-added rate further amplifies the utility of human resources in innovation, reflecting this supply's supportive role.

Hypothesis 3: Financing environments significantly moderate innovation capabilities. Enterprises with higher domestic value-added rates in exports typically possess greater capital attractiveness. Supported by stable financing environments, they secure sufficient capital to foster innovation development.

4.3 Mechanism Diagram

To visually illustrate the pathway through which local value chains drive manufacturing innovation capacity, this study has developed the following mechanism diagram (see Figure 1). As depicted, DVAR enhances corporate innovation capabilities

through the moderating effects of human resource supply and optimised financing environments, alongside the knowledge spillover effects of industrial agglomeration and the optimisation of industrial structure.

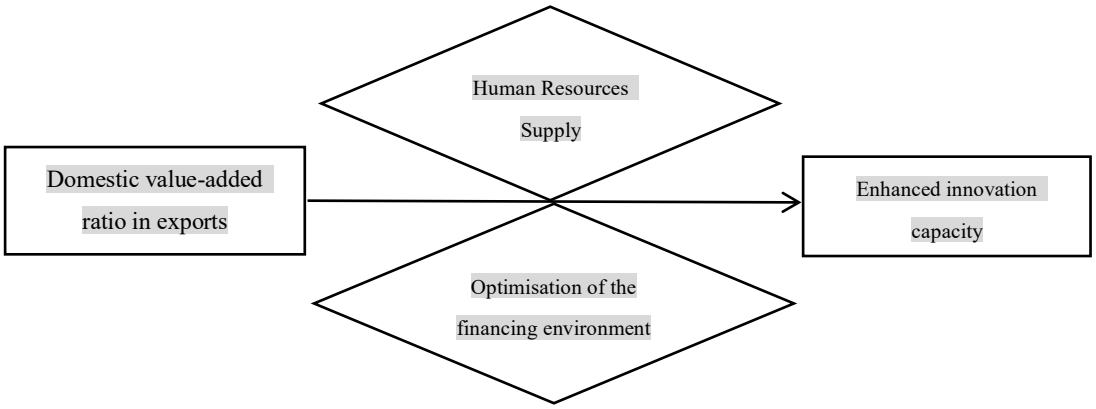

Figure 1. Mechanism diagram

The mechanism diagram illustrates how domestic value chains enhance corporate innovation capabilities through synergistic human resource supply and coordination. The core lies in enterprises increasing their domestic value-added ratio (DVAR) in exports, thereby accumulating greater resources within domestic production processes via knowledge spillovers from industrial agglomeration and structural adjustments. This forms a dynamic mechanism driving innovation.

Firstly, value chains constitute a vital foundation for corporate innovation, primarily manifesting through upstream-downstream synergies and knowledge spillover effects. Upstream-downstream synergy refers to enterprises optimising resource allocation and enhancing production efficiency by integrating resource and information flows among suppliers, manufacturers, and distributors. For instance, in manufacturing, efficient coordination between raw material suppliers, producers, and end-sales channels shortens the product lifecycle from R&D to market launch, enabling swift responses to market demands. Within this collaborative framework, enterprises secure stable resource supply while optimising production processes to reduce waste and bolster overall competitiveness.

Knowledge spillover represents another crucial manifestation of industrial chain collaboration. Through cooperation between upstream and downstream enterprises, innovative technologies can rapidly disseminate and be applied to production and management. This spillover effect is particularly vital for technology-intensive industries. For instance, technological breakthroughs by core component suppliers can significantly enhance the product performance of original equipment manufacturers, while downstream enterprises' feedback on market demands can conversely drive the technological R&D direction of upstream enterprises. This bidirectional interaction accelerates technological upgrading across the entire industrial chain, fostering the formation and expansion of regional industrial clusters.

Furthermore, the agglomeration effect within industrial chains amplifies enterprises' innovation capabilities. When a cluster of enterprises forms a geographically concentrated industrial cluster, resource sharing and technological collaboration become significantly more efficient. For instance, Silicon Valley's high-tech cluster relies not only on individual corporate innovation but also benefits from frequent interactions and technological exchanges among regional enterprises. This clustering effect creates fertile ground for innovation by reducing transaction costs and technological dissemination expenses between enterprises.

Optimising industrial structure represents another crucial mechanism within industrial chains that cannot be overlooked. As the proportion and interconnections of different industries within economic activity, industrial structure directly impacts the operational efficiency and innovation capacity of industrial chains. A rational industrial structure facilitates the coordinated development of upstream and downstream enterprises through optimised resource allocation and efficient factor mobility. For instance, when the industrial structure shifts from labour-intensive to technology-intensive sectors, greater resources and policy support are channelled towards high-value-added R&D and advanced manufacturing segments, propelling the entire industrial chain towards higher-end applications. Optimisation further enhances enterprises' capacity for technological innovation by

directing capital and labour towards high-efficiency domains. This process may gradually phase out traditional low-value-added production segments while strengthening high-value-added R&D and design functions. Such structural changes not only elevate the overall value of the industrial chain but also unlock greater innovation potential for enterprises. For instance, the advancement of high-end manufacturing often relies on advanced materials and intelligent equipment, whose resource allocation necessitates support from industrial restructuring. Within a rational industrial structure, the refinement of upstream high-tech supply chains can significantly shorten technology diffusion cycles, while downstream market demand feedback can be swiftly translated into new R&D directions.

Secondly, within the regulatory framework, human resource supply is regarded as the core driver of corporate technological innovation and management transformation. The concentration of high-end talent not only directly enhances an enterprise's R&D capabilities but also creates greater innovation opportunities both internally and externally through knowledge transfer and sharing. In modern economies, particularly within knowledge-intensive industries, the quality and quantity of talent often determine an enterprise's competitiveness and position at the technological frontier. For instance, innovation in the field of artificial intelligence is highly dependent on the contributions of top scientists and engineers, whose accumulation requires sustained resource investment and effective incentive mechanisms from enterprises.

The role of human resource supply is further evident in technology transfer and diffusion processes. The movement of high-calibre talent between enterprises and institutions accelerates knowledge dissemination and optimises the allocation of innovation resources. For instance, many technology firms collaborate with universities and research institutions, not only attracting leading researchers but also leveraging these partnerships to commercialise laboratory breakthroughs. Concurrently, internal knowledge management and training systems enable employees to rapidly master new technologies, enhancing overall R&D efficiency.

Moreover, the supply of human resources can also facilitate enterprises' global technology acquisition and competitiveness enhancement. In an internationalized context, multinational corporations recruit high-end talent globally, bringing advanced technologies and management expertise to local markets. For instance, multinational enterprises establishing R&D centers in China not only cultivate local engineers but also accelerate the localization of global technologies through collaborations with local universities and research institutions. In this way, human resource supply constructs innovation networks for enterprises within a global perspective, enabling them to gain advantages in both technological and market competition.

Simultaneously, the optimization of the financing environment lies in providing ample capital support for corporate innovation activities. Technological innovation often requires substantial R&D investment and long-term financial backing, which traditional self-funded accumulation struggles to meet. Diverse financing channels within the capital market—including capital markets, bank credit, and venture capital—inject fresh vitality into corporate innovation. For instance, startups typically rely on venture capital to fund their early R&D and market expansion phases, while established enterprises leverage capital markets financing to drive large-scale technological upgrades and industrial expansion.

The importance of financing environment optimization in corporate innovation also manifests in capital liquidity and accessibility. In technology-intensive industries, companies require continuous capital investment for R&D, and funding shortages can directly lead to the interruption or delay of innovation projects. Therefore, optimizing the financing environment must not only provide funding support but also enhance capital utilization efficiency by improving capital allocation and reducing financing costs. For instance, the recent development of green finance has provided specialized funding for technology R&D in new energy enterprises. Through green bonds and green loans, it has accelerated the application and promotion of relevant technologies.

Additionally, an optimized financing environment strengthens risk-sharing mechanisms. During technological innovation, enterprises face significant uncertainty, particularly in immature markets. Financial institutions can help mitigate the financial pressure of innovation failures through diversified investments and risk management tools. For instance, governments establish industrial guidance funds and technology subsidies, not only reducing financing costs but also providing policy safeguards for technological innovation in high-risk sectors.

Finally, innovation capability serves as the core link throughout the entire mechanism, spanning the entire process from R&D and technology transfer to market application. It tightly integrates scientific research activities, production manufacturing, and market feedback, forming a closed-loop system that drives technological breakthroughs and enhances economic benefits. By integrating internal resources and external collaborations, enterprises continuously optimize their technological capabilities and product performance, thereby strengthening their market competitiveness.

Within this enhancement of innovation capacity, technological R&D serves as the starting point. By combining fundamental and applied research, enterprises can develop new technologies and products. This phase typically requires long-term investment and interdisciplinary collaboration. For instance, new drug development in the biopharmaceutical industry involves years of exploration from molecular screening to clinical trials, necessitating not only the company's own R&D capabilities but also partnerships with universities, research institutions, and clinical hospitals.

Technology commercialization is a critical link in enhancing innovation capabilities. The outcomes of R&D must be transformed into practical products or services through industrialization. This process involves not only adapting technologies and adjusting production processes but also accurately grasping market demands. For instance, smartphone development requires integrating the latest chip, display, and communication technologies while continuously optimizing product design and feature configurations based on consumer preferences.

Market application provides a feedback mechanism. By analyzing market demands and user feedback, enterprises continuously refine existing technologies and optimize product performance. This market-driven innovation model enables companies to swiftly adapt to technological shifts and competitive landscapes while guiding future R&D directions. For instance, the electric vehicle industry leverages user data collection and analysis to advance battery technology and autonomous driving algorithms, propelling sector-wide technological progress.

The regulatory mechanism tightly integrates local value chains, human resources, financing environments, and innovation capabilities, forming a systemic driving force that propels corporate innovation capacity. The industrial chain provides the foundation for resource allocation and collaborative partnerships. The talent chain enhances technological innovation capabilities through the aggregation of high-end human capital and knowledge dissemination. The financing environment resolves funding challenges in innovation activities through capital support, creating a closed-loop system that drives technological breakthroughs. This not only strengthens corporate competitiveness but also provides crucial support for regional economic and industrial technological upgrading. The successful operation of this mechanism is key to modern enterprises gaining an edge in global competition.

4.4 The Tesla Case

In 2019, Tesla established its first wholly foreign-owned Gigafactory in Shanghai's Lingang area, becoming a model of China's manufacturing sector deeply embedded in the global value chain. This project not only demonstrates China's steadfast commitment to openness and cooperation in the new energy vehicle sector but also provides a concrete and illustrative case for studying how the domestic value-added ratio in exports (DVAR) promotes corporate innovation capabilities. By analyzing the spillover effects of Tesla's Shanghai factory on China's new energy vehicle supply chain, we can gain deeper insights into how DVAR stimulates corporate innovation motivation and capacity under the moderating effects of optimized human resource supply and financing environments.

The establishment of Tesla's Shanghai factory was remarkably swift, achieving production within just ten months from signing the agreement. This feat has been hailed as emblematic of the ‘China Speed’. Behind this rapid pace lies comprehensive support from China across industrial policy, administrative efficiency, and land supply.

As a pivotal node in the multinational corporation's global network, the Shanghai facility not only serves the domestic Chinese market but also exports substantial volumes to European and Asian markets. It thus fulfils a dual role within the global new energy vehicle value chain: manufacturing and technology export.

Concurrently, Tesla places significant emphasis on its localisation strategy. By 2023, the Shanghai facility achieved over

95% localisation of components, with enterprises including CATL, Baosteel, and Jingjin Electric becoming core suppliers. This initiative directly enhances the domestic value-added ratio within exported products.

Tesla's entry into the Chinese market represents not merely a shift in production footprint, but a localised expansion of its technological capabilities. The company has partially relocated key R&D functions, such as battery management systems (BMS), to Shanghai, strengthening domestic R&D teams' capabilities in core algorithms, materials management, and system integration.

This transfer not only increases the proportion of Chinese technology in exported vehicles but also generates tangible technology spillover effects: on one hand, Tesla's opening of certain patents provides domestic enterprises with a window for technological learning. Local manufacturers like NIO and BYD have benefited to varying degrees from Tesla in design philosophy and technical implementation. For instance, NIO's ET7 employs a module-free (CTC) battery structure, achieving a 15% increase in energy density. On the other hand, through engineer mobility and collaborative R&D platforms, Tesla has established a mechanism for the spillover of tacit knowledge. Core personnel from its former autonomous driving team have joined new-energy vehicle manufacturers like Xpeng and Li Auto, playing a pivotal role in advancing their navigation-assisted driving systems (NGP).

Furthermore, leveraging the industrial clustering effect driven by Tesla's establishment, the Lingang area has progressively formed an integrated smart electric vehicle cluster encompassing R&D, manufacturing, and testing. By 2023, this region had attracted over 200 upstream and downstream supporting enterprises. AI chip companies such as Horizon Robotics and Black Sesame Intelligence have also actively participated in collaborative innovation, elevating the overall technological sophistication and R&D capabilities of the industry. Statistical data indicates that in 2023, patent applications within this industrial cluster increased by 40% year-on-year, with invention patents accounting for a substantial 65% share. This demonstrates that technological spillover has significantly elevated enterprises' capacity for independent innovation.

Tesla's impact on China's industrial chain extends beyond the growth of a single enterprise, driving systemic optimisation of the entire industrial structure and technological capabilities. Upstream, to meet its demand for high-performance batteries, enterprises like CATL and Rongbai Technology accelerated R&D into high-nickel ternary cathode materials, thereby breaking the monopoly held by Japan and South Korea and achieving a leap in global market share. Midstream, the Shanghai factory's introduction of fully automated welding lines spurred rapid development among domestic equipment suppliers like Lead Intelligent, leading to the creation of more cost-effective and efficient domestic automation solutions. Downstream, Tesla's adoption of open Supercharger standards facilitated joint charging network development with State Grid and Telai, significantly enhancing nationwide charging infrastructure coverage and efficiency. From 2019 to 2023, China's charging pile count surged from 1.2 million to 6.8 million, with charging efficiency improving by 50%, laying a robust foundation for new energy vehicle adoption.

The positive impact of DVAR on innovation capabilities is significantly amplified by human resource support. Tesla's presence has catalysed the aggregation and cultivation of high-end talent. Through collaborations with institutions such as Shanghai Jiao Tong University and Tongji University, it established the ‘New Energy Vehicle Engineering Master's Programme’, which has produced 1,200 specialised engineers for domestic enterprises within three years. Sixty per cent of these graduates have joined domestic new energy vehicle manufacturers, driving an overall enhancement in the industry's technical capabilities. Concurrently, Tesla localised its global skills training standards by launching the ‘New Energy Vehicle Assembly Technician’ certification programme. Covering 20 vocational colleges nationwide, this initiative trains 5,000 individuals annually. This standardised talent mechanism enhances the skill levels of foundational workers, shortening the time required for technical personnel to progress from induction to proficiency.

The improved financing environment has provided capital support for the transformation between export value-added and innovation capabilities. Tesla has spurred a cohort of supply chain enterprises to enter the capital markets. Between 2020 and 2023, 32 enterprises within the new energy vehicle industry chain successfully completed IPOs on the STAR Market or ChiNext, raising over 80 billion yuan in total. Through capital expansion, these enterprises rapidly increased production capacity and

R&D investment. Concurrently, venture capital institutions significantly increased their investments in the new energy vehicle sector. In 2023, capital firms including Sequoia Capital and Hillhouse Capital established specialised funds, directing a total of 120 billion yuan towards cutting-edge technologies such as autonomous driving and solid-state batteries – accounting for 70% of the industry's total financing. Policy-based financial instruments also provided stable financing channels for small and medium-sized enterprises. The People's Bank of China incorporated new energy vehicles into its carbon reduction support framework, offering green loans at an annual interest rate of 1.75% to eligible enterprises. In 2023, green loans to the new energy vehicle sector reached 200 billion yuan. This capital infusion not only addressed short-term funding needs but also bolstered enterprises' willingness and capacity for long-term technological investment. Consequently, within an optimised financing environment, DVAR has been catalysed into sustained innovation capability.

With the sustained rise in DVAR and the ample supply of innovation factors, China's new energy vehicle industry has entered a period of concentrated innovation capability. In 2023, the industry filed 128,000 patent applications, representing a 280% increase from 2019, with invention patents accounting for 58% of the total. CATL's CTP module-free battery patent has secured authorisation in 20 countries worldwide, demonstrating the growing international influence of Chinese enterprises' technological innovation.

5 Research Methodology

5.1 Data Sources and Sample Selection

This study draws upon data from the China Industrial Enterprise Database and the China Customs Database, supplemented by regional statistical yearbooks, to conduct an empirical analysis of corporate innovation capabilities and their influencing factors. Details are as follows:

1. Domestic Value Added Ratio of Exports (DVAR) is calculated with the methodology proposed by Lv Yue et al. (2018)[†], derived from matched customs and industrial enterprise database records.

2. Innovation Capacity (TCIN_log) is measured by the logarithm of the number of patent authorisations granted to enterprises, sourced from the China Patent Database.

3. Human resources (PEOPLE) reflects the supply of human resources through annual enterprise employment figures sourced from the industrial enterprise database.

4. Financing environment (FE) is constructed based on the indicators proposed by Qi et al. (2024), using principal component analysis to measure the financing environment[‡] in which enterprises operate.

5. Control variables include firm age (AGE), total assets (TA), debt-to-asset ratio (DAR), and total asset growth rate (TAGR). These variables are employed to control for the potential influence of firm characteristics on innovation capability.

Table 1. Summary Table of Control Variables

Type of variable	Variable name	symbol	Account form
Control variables	enterprise age	AGE	2024-Company Year of Establishment
	total assets of enterprise	TA	total assets of enterprise
	corporate debt ratio	DAR	total indebtedness /total assets
	total assets growth rate	TAGR	(total assets at end of year-total assets at end of prior year)/total assets at end of year

During the sample screening process, to avoid the interference of outliers on regression results, and the following samples were excluded:

1. Enterprises with missing values for key variables such as innovation capability and debt-to-asset ratio;

2. Enterprises where the domestic value-added rate of exports or debt-to-asset ratio exceeded reasonable ranges (less than 0 or greater than 1).

[†] Lv Yue, Sheng Bin, Lü Yunlong. Does Market Segmentation in China Lead to a Decline in the Domestic Value-Added Rate of Exporting Enterprises? [J]. China Industrial Economics, 2018(05):5-23.

[‡] The principal component analysis method is detailed in Appendix 1.

The final panel data comprises 219,095 manufacturing enterprises, covering the period from 2000 to 2014.

5.2 Variable Definitions and Measurement

1. Dependent variable: Innovation capacity (TCIN_log), measured using the logarithm of patent counts to gauge firms' technological innovation output.

2. Core Explanatory Variable: Domestic Value Added Ratio of Exports (DVAR), reflecting the local value contribution of a firm's products.

Interaction Variables:

- DVAR × PEOPLE: To examine the moderating effect of human resource supply on the relationship between value added ratio and innovation capability;
- DVAR × FE: To examine the moderating effect of optimised financing conditions on the relationship between value added ratio and innovation capability.

3. Control variables:

- Age (AGE): Years since establishment, reflecting accumulated resources;
- Total assets (TA): Measured in thousands of yuan, indicating firm size;
- Debt-to-asset ratio (DAR): Reflecting financial leverage;
- Total asset growth rate (TAGR): Measuring firm growth potential.

5.3 Empirical Model Specification

To examine the mechanism through which a firm's export domestic value-added ratio influences manufacturing innovation capability, this study employs the following fixed-effects model:

$$\mathbf{TCIN_{it} = \alpha_0 + \alpha_1\ DVAR + \alpha_2\ DVAR_{it} * PEOPLE_{it} + \alpha_3\ DVAR_{it} * FE_{it} + \alpha_4\ CONTROLS_{it} + \mu_i + \varepsilon_{it}}$$

Where:

- $\mathbf{TCIN_{it}}$: Innovation capability of firm i in year t;
- $\mathbf{DVAR_{it}}$: Firm's export domestic value-added ratio;
- $\mathbf{PEOPLE_{it}}$: Firm's human resource supply;
- $\mathbf{FE_{it}}$: Firm's financing environment indicator;
- $\mathbf{CONTROLS_{it}}$: Control variables, including age, total assets, debt-to-asset ratio, and total asset growth rate;
- μ_i : Firm-specific fixed effect;
- ε_{it} : error term.

5.4 Descriptive Statistics

To gain an intuitive understanding of the fundamental characteristics of the core variables, descriptive statistics were applied to the data in this study. The specific results are as follows:

Table 2. Descriptive Statistics

Variable name	variable symbol	observed value	mean value	standard deviation	minimum	maximum
Domestic value-added ratio in exports	DVAR	219095	0.83	0.26	0	1
Number of patents	TCIN	219095	1.89	28.91	0	6122
human resources,	PEOPLE	219095	467.85	1152.11	5	96537
financing environment	FE	219095	0.03	0.43	-0.67	23.08
Year of establishment	AGE	219095	26.35	7.56	13	187
Total Assets (In Thousands of CNY)	TA	219095	234284.61	1771281.49	400	189407432.00
Debt-to-Asset Ratio	DAR	219095	0.52	0.24	0	1
Total Assets Growth Rate	TAGR	219095	0.05	0.35	-0.5	1

According to the descriptive statistics in Table 2, the sample data encompasses 219,095 observations. The domestic value-added ratio (DVAR) in exports is 0.83, with a standard deviation of 0.26, a minimum value of 0, and a maximum value of 1. This indicates that the export value-added rates of most enterprises are relatively concentrated, though some variability persists. The mean for the number of patents (TCIN) is 1.89, with a standard deviation of 28.91, a minimum value of 0, and a maximum value of 6122. This reveals an uneven distribution of patent holdings, with some enterprises possessing a large number of patents while the majority hold relatively few. The mean for human resources (PEOPLE) is 467.85, with a standard deviation of 1152.11, minimum value of 5, and maximum value of 96537. This indicates significant variation in workforce size across enterprises, with some possessing exceptionally large human resource pools. The mean for financing environment (FE) is 0.03, with a standard deviation of 0.43, ranging from a minimum of -0.67 to a maximum of 23.08. This indicates substantial volatility in financing environment data and significant disparities in financing conditions across enterprises. The mean for years since establishment (AGE) is 26.35 years, with a standard deviation of 7.56 years, a minimum of 13 years, and a maximum of 187 years. This reflects a broad age range among the sampled enterprises, encompassing both long-established companies and relatively young firms. The mean value for total assets (in thousands of CNY) (TA) is 234,284.61 thousand yuan, with a standard deviation of 1,771,281.49 thousand yuan, a minimum value of 400 thousand yuan, and a maximum value of 189,407,432 thousand yuan. This indicates highly significant differences in the scale of corporate assets. The mean debt-to-asset ratio (DAR) was 0.52, with a standard deviation of 0.24, a minimum value of 0, and a maximum value of 1. This reflects that most enterprises have moderate debt levels, though extreme cases exist. The mean of the Total Asset Growth Rate (TAGR) is 0.05, with a standard deviation of 0.35, a minimum value of -0.5, and a maximum value of 1. This indicates substantial variation in total asset growth among enterprises, with some experiencing significant asset expansion while others potentially facing asset contraction.

5.5 Correlation Analysis

Table 3 indicates that the domestic value-added ratio in exports (DVAR) exhibits a significant positive correlation with both innovation capability (TCIN) and financing environment (FE), with correlation coefficients of 0.01 and 0.09 respectively. Both correlations are statistically significant at the 1% level, preliminarily validating the hypothesis that the domestic value-added rate of exports positively promotes innovation capability. Specifically, a higher domestic value-added ratio indicates an enhanced local capability. This strengthened capability enables enterprises to access greater resource allocation, including increased technological support and financing capacity, thereby promoting improvements in their innovation capabilities. Furthermore, the domestic value-added ratio in exports (DVAR) exhibits a significant positive correlation with the debt-to-asset ratio (DAR) (correlation coefficient 0.08, $p < 0.01$), further supporting its positive impact on corporate capital attractiveness and providing evidence for the supportive role of the financing environment.

The correlation coefficient between the number of employees (PEOPLE) and total assets (TA) is 0.51, significant at the 1% level, indicating a strong positive relationship between enterprise scale and resource base. Simultaneously, the number of employees (PEOPLE) is negatively correlated with the domestic value-added ratio in exports (DVAR) (correlation coefficient: -0.06, $p < 0.01$). This phenomenon may reflect that higher value-added rates are concentrated in innovation-driven enterprises, which may operate with leaner staffing. Nevertheless, the increase in value-added rate still enhances their innovation capacity through optimized human resource allocation and concentrated investment.

Combined with Hypothesis 2 mentioned earlier, this phenomenon demonstrates the moderating effect of the talent chain: as the value-added rate rises, the utility of human resources in innovation activities significantly increases.

The moderating effect of financing environment stability on innovation capability is also evident in Table 1. The correlation coefficient between financing environment (FE) and debt-to-asset ratio (DAR) is 0.23, significant at the 1% level, indicating that enterprises with sound financial health possess stronger capabilities in capital operations. The positive correlation between the domestic value-added ratio in exports (DVAR) and financial leverage (DAR) (correlation coefficient

0.08, p < 0.01) indicates that enterprises with high value-added rates often possess stronger capital attractiveness. They can secure greater financial support under an optimized financing environment, thereby promoting the development of innovation capabilities. This result preliminarily validates Hypothesis Three, namely the important moderating role of the financing environment in enhancing corporate innovation capabilities.

Regarding the maximum correlation coefficient among variables, the correlation between enterprise size (PEOPLE) and total assets (TA) was 0.51, below 0.6. This suggests a low likelihood of multicollinearity among variables in the model, making it suitable for further regression analysis. In summary, the correlation analysis results in Table 1 provide preliminary statistical support for the subsequent empirical research in this paper, validating the theoretical relationships among some variables.

Table 3. Correlation Analysis

	DVAR	TCIN	FE	PEOPLE	AGE	TA	DAR	TAGR
DVAR	1							
TCIN	0.01***	1						
FE	0.09***	-0.01**	1					
PEOPLE	-0.06***	0.36***	-0.02***	1				
AGE	-0.02***	0.02***	-0. 12***	0.09***	1			
TA	-0.04***	0.39***	-0.02***	0.51***	0.05***	1		
DAR	0.08***	0.01***	0.23***	0.02***	-0.04***	0.02***	1	
TAGR	-0.01***	-0.02***	0.07***	-0.08***	-0.04***	-0.05***	-0.03***	1

5.6 Baseline Regression Analysis

To examine how the domestic value-added ratio in corporate exports influences manufacturing innovation capacity using a fixed-effects model, the following baseline regression results are presented:

Table 4. Fixed-effects Regression Results

	TCIN_log	TCIN_log	TCIN_log
DVAR	0.1624***	0.421***	0.035***
	（25.57）	（4.04）	(3.12)
PEOPLE	0.00004***	0.00005***	0.00003***
	（11.75）	（7.65）	(1.78)
FE	0.5049*	-0.1124	0.0069
			(0.61)
AGE	（3.54）	（-0.69）	-0.0036***
			(-3.13)
TA			0.00000***
			(4.98)
DAR			-0.097***
			(-9.18)
TAGR			-0.086***
			(-25.2)
DVAR*PEOPLE	0.00014***	0.00006***	0.0001***
	（29.97）	（7.80）	(3.15)
DVAR*FE	-0.1347*	-0.00126	-0.0115
	（-8.57）	（-0.07）	(-0.94)
Individual fixed effect	NO	YES	YES
Time fixed effect	NO	NO	NO
sample capacity	219095	219095	219095
R square	0.0662	0.49	0.6296

Notes: ***p<0.001, **p<0.01, *p<0.05

The regression results reveal:

1. Core variables: The regression coefficient for the domestic value-added ratio in exports (DVAR) is 0.035, significant at the 1% level. This indicates that an increase in DVAR significantly enhances an enterprise's innovation capacity, consistent with Hypothesis 1.

2. Interaction terms: The coefficient for DVAR*PEOPLE is 0.0001, significantly positive, indicating that the number of employees enhances the promotional effect of the value-added rate on innovation capacity, thus validating Hypothesis 2. The coefficient for DVAR*FE is -0.011 and not significant, suggesting that the moderating effect of the financing environment on the relationship between the value-added rate and innovation capacity is insignificant. This may be related to the persistence of financing constraints for some enterprises.

3. Control variables: Firm Age (AGE), debt-to-asset ratio (DAR), and total asset growth rate (TAGR) all exerted significant negative effects on innovation capability. This may reflect mature firms' conservative tendencies towards innovation, alongside the constraints imposed by high debt levels and asset growth on innovative activities.

5.7 Robustness Tests

To validate the reliability of the benchmark regression results, robustness tests were conducted. A substitution method for the dependent variable was employed for verification. The metric for innovation capability—originally measured by total patent count—was replaced with the count of utility model patents. Utility model patents represent a patent category with practical application value, typically reflecting more direct technological innovation outcomes when assessing corporate innovation capability. This substitution allows us to examine whether the original conclusions depend on the choice of patent type and to validate the effectiveness of utility model patents in measuring innovation capability.

Table 5. Robustness Test

	TCIN2_log
DVAR	0.29***
	(2.65)
PEOPLE	0.00001***
	(0.65)
AGE	0.00104***
	(-2.21)
FE	-0.00284
	(0.016)
TA	0.00000***
	(5.41)
DAR	-0.0752***
	(-9.11)
TAGR	-0.599***
	(-22.7)
DVAR*PEOPLE	0.00011***
	(2.68)
DVAR*FE	-0.00076
	(-0.09)
individual determinism effect	YES
Time-fixed effect	NO
sample capacity	219095
R square	0.51

Notes: ***p<0.001, **p<0.01, *p<0.05

The regression results indicate that when the logarithm of utility patent usage is employed as the dependent variable, the regression coefficient for DVAR remains significantly positive and consistent with the benchmark regression findings. Specifically, the coefficient for DVAR is 0.29, which remains statistically significant at the 1% level. This demonstrates that the positive impact of the value-added rate on innovation capability holds not only in the benchmark regression but is also

validated when using alternative measures of innovation capability.

5.8 Heterogeneity Analysis

To further examine the moderating effect of industry characteristics on the relationship between value-added rate and innovation capability, we subdivided industries based on the proportion of labor, capital, and technology factors, following Xu Jianrong (2011). This categorization divides manufacturing into three major types: capital-intensive, labor-intensive, and technology-intensive, as shown in the table:

Table 6. Manufacturing Classification Based on Factor Intensity

Industry type	**Included Sub-industries**
Labor-intensive	1 Processing of Agricultural and Sideline Products, 2 Food Manufacturing, 5 Textile Industry, 6 Textile, Apparel, and Accessories Industry, 7 Leather, Fur, Feather, and Related Products and Footwear Manufacturing, 8 Wood Processing and Wood, Bamboo, Rattan, Palm, and Grass Products Industry, 9 Furniture Manufacturing, 11 Printing and Reproduction of Recorded Media, 12 Manufacture of Educational, Artistic, Sporting, and Entertainment Goods, 17 Rubber and Plastic Products Industry, 18 Non-metallic Mineral Products Industry, 21 Metal Products Industry
Technology-intensive	15 Pharmaceutical Manufacturing 23 Special-purpose Equipment Manufacturing 25 Railway, Shipbuilding, Aerospace, and Other Transportation Equipment Manufacturing 26 Electrical Machinery and Equipment Manufacturing 27 Computer, Communication, and Other Electronic Equipment Manufacturing 28 Instrument and Meter Manufacturing
Capital-intensive	3 Alcoholic Beverages, Soft Drinks, and Refined Tea Manufacturing 4 Tobacco Products Manufacturing 10 Paper and Paper Products Manufacturing 13 Petroleum, Coal, and Other Fuel Processing 14 Chemical Raw Materials and Chemical Products Manufacturing 16 Chemical Fiber Manufacturing 19 Ferrous Metal Smelting and Rolling 20 Non-Ferrous Metal Smelting and Rolling 22 General Equipment Manufacturing

The regression results are as follows:

Table 7. Regression Results Distinguishing Labor-intensive, Technology-intensive, and Capital-intensive Industries

	Labor-intensive	TCIN_log Technology-intensive	Capital-intensive
DVAR	-0.023* (0.010)	0.080*** （0.018）	0.006 （0.024）
PEOPLE	0.00000 (0.00001)	0.00002** （0.00001）	0.00004** （0.00002）
AGE	-0.003** (0.001)	-0.005*** （0.002）	-0.027 （0.041）
FE	0.030 (0.0018)	-0.029 （0.032）	-0.001 （0.003）
TA	0.00000*** （0.000）	0.00000*** （0.000）	0.00000*** （0.000）
DAR	-0.024* (0.0013)	-0184*** （0.021）	-1.04*** （0.029）
TAGR	-0.034*** (0.005)	-1.37*** （0.007）	-0.065*** （0.010）
DVAR*PEOPLE	0.0001*** (0.00001)	0.0001*** (0.00001)	-0.00001 (0.00002)
DVAR*FE	-0.024 (0.020)	0.011 （0.034）	0.043 （0.045）
Individual Fixed Effects	YES	YES	YES
sample capacity	219095	219095	219095
R square	0.443	0.694	0.3565

Notes: ***p<0.001, **p<0.01, *p<0.05

The estimation results from the heterogeneity analysis reveal that the core explanatory variable DVAR examined in this study exhibits significant differences in its impact across labor-intensive, technology-intensive, and capital-intensive industries. Specifically, the regression coefficient for DVAR is significantly positive in technology-intensive industries, indicating that the export domestic value-added rate in this sector can substantially enhance innovation capabilities. In labor-intensive industries, however, the regression coefficient for DVAR is negative, suggesting that an increase in export value-added rate may exert a certain inhibitory effect on innovation capacity. For capital-intensive industries, the regression coefficient for DVAR is close to zero and not significant, indicating no significant linear relationship between export value-added rate and innovation capacity in this sector.

Comparing regression coefficients across industry types reveals significant sectoral heterogeneity in DVAR's impact on innovation capacity. This confirms that the effect of export value-added rate on innovation varies by industry characteristics, with technology-intensive sectors exhibiting the strongest DVAR-driven innovation promotion. Conversely, labor-intensive and capital-intensive sectors show weaker effects, with labor-intensive industries even displaying a negative relationship. This sectoral divergence likely stems from differences in technological sophistication, capital requirements, and innovation resources across industries.

Furthermore, the significant positive effect of the interaction term DVAR*PEOPLE in technology-intensive and labor-intensive sectors further indicates that both export value-added rate and increased employment jointly drive innovation capacity enhancement. However, the interaction term is insignificant in capital-intensive industries, likely because these sectors have lower labor demands, with innovation driven more by capital and technological accumulation than by human resource growth. Meanwhile, DVAR*FE exhibits low significance across all three intensity sectors, reflecting the limited contribution of China's manufacturing financial environment to fostering innovation capabilities and its consequently poor effectiveness within the mechanism.

5.9 Endogeneity Test

Considering the potential bidirectional causality between the dependent variable (innovation capability) and the independent variable (DVAR)—where the value-added rate (DVAR) may not only influence innovation capability but also be affected by it in return—this study employs DVAR lagged by one period as an instrumental variable for 2SLS (two-stage least squares) estimation. The regression results are as follows:

Table 8. Endogeneity Test

	（1）	（2）
	TCIN_log	TCIN_log
Lag_DVAR	0.7510***	
	(483.94)	
DVAR		0.2684***
		(26.84)
AGE	-0.00027***	0.0009***
	(-3.41)	(2.35)
TA	0.00000***	0.00000***
	(-6.32)	（83.63）
DAR	0.0164***	0.5077***
	(9.5)	（6.05）
TAGR	-0.0498***	-0.2055***
	(-2.60)	（-22.13）
Individual Fixed Effects	YES	YES
Time Fixed Effects	NO	NO
Underidentification test		0.0095
Weak identification test		23.05
Sargan statistic		0.0000

Notes: ***p<0.001, **p<0.01, *p<0.05

In the first-stage regression, the coefficient of the instrumental variable Lag_DVAR is 0.7510, significant at the 1% level. The instrumental variable passes both the underidentification test and the weak instrumental variable test, further confirming its validity. Specifically, the p-value for the underidentification test was 0.0095, less than 0.05, indicating the model passed the underidentification test. The instrumental variable effectively identified the endogenous independent variable, rendering the model identifiable. Additionally, the statistic for the weak identification test was 23.05, greater than 10, confirming the instrumental variable possessed sufficient correlation and did not suffer from weak instrumentation issues.

In the second-stage regression, the table shows that the coefficient for DVAR is 0.2684 and is significant at the 1% level. This indicates that even after controlling for endogeneity issues, the positive impact of the value-added rate on innovation capability remains significant. This result further validates the robustness of the model, demonstrating that after eliminating endogeneity bias, the value-added rate (DVAR) still significantly promotes innovation capability.

In summary, this study resolves potential endogeneity issues by introducing the lagged value-added rate as an instrumental variable, with its validity confirmed through relevant tests. The two-stage estimation results demonstrate that the model remains robust after addressing endogeneity, proving that the relationship between DVAR and innovation capability is stable.

6 Conclusions and Policy Recommendations

6.1 Research Findings

This study explores the impact of DVAR on innovation capabilities among Chinese manufacturing enterprises, focusing on the driving role of the value-added rate in enhancing corporate innovation capacity. It further examines the moderating effects of human resource supply and financing environment optimization. Through empirical analysis of Chinese manufacturing enterprise data, the study draws the following key conclusions:

The empirical results demonstrate a significant positive relationship between a firm's domestic value-added ratio (DVAR) in exports and its innovation capability. This finding supports Schumpeterian innovation theory, which posits that firms enhance their innovation capability by increasing the value-added of their technologies and products, thereby improving their market competitiveness.

Heterogeneity analysis reveals significant differences across industries in how the value-added ratio influences innovation capability. Firms in technology-intensive industries exhibit higher sensitivity to value-added rates, whereas labor-intensive and capital-intensive sectors demonstrate weaker correlations.

When analyzing their combined effects, the study indicates that the moderating effect of DVAR on human resource supply can optimize mechanism chain links by facilitating technological R&D through talent acquisition. This enhances technological innovation efficiency and market adaptability, effectively boosting manufacturing firms' innovation capabilities. Optimized financing environments provide capital support for innovation activities; talent chains facilitate this process; and enhanced innovation capabilities drive a virtuous cycle across the entire innovation ecosystem through the commercialization of technological achievements. However, despite financing environments being crucial for innovation activities, this study did not find a significant moderating effect on the relationship between value-added rate and innovation capability. This may be attributed to persistent financing constraints in some enterprises, indicating limited alignment between financing environments and value chains.

In summary, the core conclusion of this study is that the domestic value-added ratio in exports exerts a significant positive influence on innovation capabilities, with this effect exhibiting heterogeneity across different types of enterprises and regions. By optimizing human resource supply and enhancing the financing environment to facilitate value chain regulation, manufacturing enterprises can elevate their innovation capabilities and secure a more competitive position in the global market.

6.2 Policy Recommendations

This study reveals the positive impact of the domestic value-added ratio in exports (DVAR) on the innovation capacity of manufacturing enterprises within the context of global value chain integration, while highlighting the moderating roles of human resource supply and financing environment optimization. Based on empirical findings and considering the current challenges facing China's manufacturing sector—such as technological barriers, rising labor costs, and regional development imbalances—the following policy recommendations are proposed to advance the sector's transition toward high-end, intelligent, and green manufacturing, thereby achieving high-quality development.

First, the government should implement stronger policy incentives to encourage enterprises to increase R&D investment, particularly through tax incentives, fiscal subsidies, and R&D funding support. This will drive enterprises to enhance product value-added and foster technological innovation. Simultaneously, the state should prioritize support for technology-intensive industries, helping them achieve breakthroughs in R&D and innovation activities, thereby elevating the overall technological capability and global competitiveness of the manufacturing sector.

Financial support is undoubtedly a critical component in the process of manufacturing innovation. The government must continuously refine the financial support system, especially by strengthening financing services for small and medium-sized enterprises (SMEs). Specific measures could include streamlining financing procedures, lowering loan interest rates, and expanding dedicated funds for scientific and technological innovation. For enterprises with strong technological innovation capabilities, more flexible and diverse financing tools—such as equity financing and bond issuance—should be provided to ensure adequate capital support for sustained innovation activities.

Similarly, talent is the core driver of innovation. Governments should formulate policies to attract and retain high-end innovative talent, particularly professionals engaged in R&D. Specific measures include establishing dedicated talent recruitment programs, optimizing the talent development environment, and providing research funding. Furthermore, governments should intensify training for highly skilled labor, encouraging manufacturing enterprises to collaborate with universities and research institutions to ensure access to essential technical and managerial talent, thereby enhancing overall innovation capabilities.

Second, governments should guide the optimization and restructuring of industrial chains, particularly by strengthening collaboration between upstream and downstream enterprises to promote the deep integration of information flows, technology flows, and resource flows. As the digital economy increasingly becomes a dominant force, manufacturing enterprises should leverage cutting-edge technologies such as big data and artificial intelligence to enhance every link in the industrial chain, further improving production efficiency and innovation capabilities. Governments should also encourage cross-industry cooperation, support technological innovation and cross-sector applications across different industries, and drive the overall advancement of technological standards throughout the manufacturing sector.

Finally, given regional disparities in economic development, governments should strengthen policy coordination across regions, particularly by boosting manufacturing innovation capacity in central and western China. Measures such as expanding infrastructure investment, optimizing innovation environments, and facilitating technology transfer can narrow regional innovation gaps. Concurrently, governments can provide financial support for innovation activities in these regions through fiscal transfers, thereby driving technological innovation and value-added growth among local enterprises.

6.3 Limitations of This Study

Although this study offers valuable insights into the relationship between export domestic value-added rate and innovation capacity, several limitations exist. Future research may expand in the following areas:

1. The sample data primarily originates from Chinese manufacturing enterprises and spans a relatively short timeframe. Future studies should consider broadening the industry coverage of the sample and incorporating enterprise data from more regions to enhance the universality and credibility of conclusions. Additionally, employing more panel data models could facilitate deeper analysis of potential time and regional effects, thereby improving the robustness of findings.

2. While this study examines variables such as value-added rate, innovation capacity, human resources, and financing environment, the selection of indicators for measuring innovation capacity may be limited. Future research could adopt more multidimensional innovation capacity metrics, such as patent quality and R&D input-output ratios, to enrich measurement and analytical approaches.

3. Although this study addressed endogeneity issues using instrumental variables, unaccounted endogenous factors may still exist. Future research could explore more advanced techniques like System Generalized Method of Moments (System GMM) or Dynamic Panel Data Models to enhance estimation accuracy and conclusion reliability.

4. Future research may consider cross-country comparisons, particularly examining how manufacturing firms in different nations enhance innovation capabilities by increasing value-added rates. Such global comparisons will deepen understanding of how global value chains influence innovation capacity and reveal differences among countries in industrial chain optimization and technological innovation.

5. Future research should also delve into the specific implementation pathways and effectiveness evaluation of government policies in practice, particularly practical experiences in promoting corporate innovation through policy regulation. Exploring the effectiveness of policy instruments, implementation challenges, and optimization pathways will provide valuable insights for subsequent policy formulation.

References

Chen Litian, Zhao Xiaoqing, Wei Zhishan. The Essence and Evolution of Corporate Innovation Capability: A Systematic Literature Review [J]. Science and Technology Progress and Policy, 2012, 29(14): 154-160.

Chen Xionghui, Chen Mingcong, Sun Xihuan, Luo Wei, Lü Chuntian. Evaluation Study on the Integration Level of “Four Chains” [J]. China Science and Technology Forum, 2021(07): 107-114. Four Chains" Integration Development Level Evaluation Research: A Case Study of Guangdong Region [J]. China Science and Technology Forum, 2021(07):107-114.

Dong Youde, Meng Xing. OFDI, Reverse Technology Spillovers, and Domestic Enterprise Innovation Capability: An Empirical Test Based on China's Value Chain Data [J]. International Trade Issues, 2014(09):120-129.

Gao Qiang, Song Lin. Regional Innovation, Intermediate Goods, and Export Domestic Value-Added Rate [J]. International Trade Issues, 2022, (04): 158-174.

Gao Xiang, Liu Qiren, Huang Jianzhong. Factor Market Distortions and Chinese Enterprises' Export Domestic Value-Added Rate: Evidence and Mechanisms [J]. World Economy, 2018, 41(10): 26-50.

He Qiong, Qu Li. Empirical Analysis of Digital Development Level's Impact on Corporate Innovation Capability [J]. Statistics and Decision Making, 2022, 38(13): 174-178.

Liu Zhangsheng, Song Deyong, Gong Yuanyuan et al. Industry Differences and Influencing Factors of Green Technological Innovation Capability in China's Manufacturing Sector [J]. Journal of Intelligence, 2017, 36(01): 194-200.

Jiao, Yanan. Research on the Mechanism of Technological Innovation's Impact on Domestic Value Added in China's Manufacturing Exports [D]. Dongbei University of Finance and Economics, 2018.

Li, Jian; Zhang, Jinlin; Dong, Xiaofan. How the Digital Economy Affects Corporate Innovation Capabilities: Internal Mechanisms and Empirical Evidence [J]. Economic Management, 2022, 44(08): 5-22.

Li Nan. A Study on the Mechanism of Environmental Regulations Enhancing the Domestic Value-Added Rate of Enterprise Exports [D]. Northwest University. 2022.

Li Shengqi, Mao Qilin. Upstream Monopoly in Manufacturing and the Domestic Value-Added of Enterprise Exports: Empirical Evidence from China [J]. China Industrial Economics, 2017, (03):101-119.

Lv Yue, Sheng Bin, Lü Yunlong. Does Market Segmentation in China Reduce the Domestic Value-Added Rate of Exported Products? [J]. China Industrial Economics, 2018(05):5-23.

Lv Yue, Yu Xiao. Service Sector Opening, Innovation Driven Development, and Export Domestic Value Added of

Manufacturing Enterprises [J]. International Business Research, 2022, 43(03):1-17.

Liu Yuhai, Liao Sainan, Zhang Li. Tax Incentives and Export Domestic Value Added Ratio of Enterprises [J]. China Industrial Economics, 2020(09):99-117.

Liu Huizheng, Zhang Jingyi, Fang Senhui. Trade Digitalization and Domestic Value-Added Rate of Enterprise Exports [J]. International Business (Journal of the University of International Business and Economics), 2022(05):69-88.

Qiu Shilei, Wang Zilong, Yang Wankun et al. Spatial Agglomeration Effects of Innovation Capabilities in High-Tech Industries [J]. Research and Development Management, 2018, 30(06): 128-137.

Qi Ping, Song Weihui, Gao Yuanbo. Impact of the Digital Economy on the Integration of the “Four Chains” in Manufacturing [J/OL]. Contemporary Finance and Economics: 1-13 [2024-03-14 20:45]

Sha Wenbing, Li Guixiang. FDI Knowledge Spillovers, Autonomous R&D Investment, and Innovation Capability of Domestic High-Tech Enterprises: An Empirical Test Based on China's High-Tech Industry Dynamic Panel Data Model [J]. World Economy Research, 2011(05):51-56+88.

Sheng Bin, Wang Hao. Bank Branch Expansion and Domestic Value-Added Rate of Corporate Exports: A Perspective Based on the Geographic Structure of Financial Supply [J]. China Industrial Economics, 2022(02): 99-117.

[20] Sheng Chaoxun. Approaches and Strategies for Advancing the Modernization of China's Industrial Chains [J]. Reform, 2019(10): 45-56.

Song Lin, Li Ke, Gao Qiang. The Impact of Digital Technology Innovation on the Domestic Value-Added Rate of Enterprise Exports: Empirical Evidence from China's Manufacturing Sector [J]. International Trade and Economic Exploration, 2025, 41(03): 4-21.

Wu Dan, Hu Jing. Spatiotemporal Variability in National Scientific and Technological Innovation Capabilities: A Comparative Analysis of China and Ten Global Countries [J]. Science and Technology Progress and Countermeasures, 2018, 35(20): 128-136.

Wu Baiyu, Peng Benhong, Liu Jun et al. Influencing Factors of China's Manufacturing Technological Innovation Capability [J]. China Science and Technology Forum, 2016(08): 23-30.

Wang Yi, Huang Xianhai, Yu Xiao. Does Environmental Regulation Reduce the Domestic Value-Added Rate of Chinese Export Enterprises? [J]. International Trade Issues, 2019(10):117-131.

Wu Chuhao, Tang Jing. Digital Technology Competition, Infrastructure, and the Domestic Value-Added Rate of Enterprise Exports [J]. World Economy Research, 2024(03):47-63+136.

Wu Jinming, Shao Chang. Research on the Formation Mechanism of Industrial Chains: The “4+4+4” Model [J]. China Industrial Economics, 2006(04):36-43.

Wang Yudong, Zhang Bo, Wu Chuan, Xu Yulian. Measuring Synergy Between Innovation Chains and Capital Chains in High-Tech Industries: Based on a Composite System Synergy Model [J]. Science and Technology Progress and Countermeasures, 2019, 36(23): 63-68.

Wang Jifa, Ao Haiyan, Chen Hang. Research on the Chain Structure and Value Realization Mechanism of Science and Technology Service Industries Based on the Innovation Chain [J]. Science and Technology Progress and Countermeasures, 2015, 32(15): 59-63.

Xie Yumin, Zhou Kaituo, Wei Xiaoping. Study on the Reverse Spillover Effects of Foreign Direct Investment on China's Innovation Capability [J]. Economic Latitude and Longitude, 2014, 31(03): 42-47.

Xie Cong, Wang Qiang. Analysis of Spatiotemporal Pattern Evolution and Influencing Factors of Technological Innovation Capability in China's New Energy Industry [J]. Geographical Research, 2022, 41(01): 130-148.

Xu Jianrong. Research on Structural Changes in China's Manufacturing Sector During the Transition Period [D]. Nanjing University of Aeronautics and Astronautics, 2011.

Zhang Yang, Tang Lingbing, Jin Peizhen. Measurement and Influencing Factors of Innovation Capability in BRICS Countries [J]. Chinese Soft Science, 2015(06):148-157.

Zhang Jie, Liu Zhibiao, Zheng Jianghuai. Key Influencing Factors of Innovation Activities in Chinese Manufacturing Enterprises: An Analysis Based on a Questionnaire Survey of Jiangsu Manufacturing Firms [J]. Management World, 2007(06):64-74.

Zhao Shengmin, Yu Xinghui. The Impact of Corporate Social Responsibility on Enterprise Innovation: Empirical Evidence from Chinese Listed Companies [J]. Science and Technology Management, 2023, 44(04): 144-153.

Shao Chaodui, Su Danni, Li Kunwang. Service Sector Liberalization and Domestic Value-Added Rate of Enterprise Exports: Theory and Evidence from China [J]. World Economy and Finance, 2020, 43(08): 123-147. [J]. World Economy, 2020, 43(08): 123-147.

Zhao Chen, Lin Chen, Gao Zhonghua. Pathways for the Integrated Development of Innovation Chains and Industrial Chains Supported by Talent Chains: Logical Framework, China-US Comparison, and Policy Implications [J]. China Soft Science, 2023(11): 23-37.

Zhang Yingying, Bai Dongbei, Gao Yu. How Does Technology Market Development Affect the Domestic Value-Added Rate of Enterprise Exports? Empirical Evidence from China [J]. International Economic and Trade Exploration, 2020, 36(06): 25-41.

BARON D P.Managerial contracting and corporate social responsibility[J].Journal of Public Economics,2008,92(1):268-288.

BROWN J R,MARTINSSON G,PETERSEN B C.Do financing constraints matter for R&D?[J].European Economic Review,2012,56(8):1512-1529.

Fagerberg J., Lundvall, BÅ . and Srholec, M. Global Value Chains, National Innovation Systems and Economic Development[J]European Journal of Development Research,2017,30(7):557-574.

GAO F,LISIC L L,ZHANG I X.Commitment to social good and insider trading[J].Journal of Accounting and Economics,2014,57(2):149-175.

Gehl Sampath P., Vallejo, B. Trade, Global Value Chains and Upgrading: What, When and How?[J]European Journal of Development Research,2018,30(3):481-504.

Haakonsson S.J., Slepniov, D. Technology Transmission Across National Innovation Systems: The Role of Danish Suppliers in Upgrading the Wind Energy Industry in China. [J]European Journal of Development Research,2018,30(3):462-480.

HE J J,TIAN X.The dark side of analyst coverage:The case of innovation[J].Journal of Financial Economics,2013, 109(3):856-878.

Keun Leea, Marina Szapirob, and Zhuqing Mao.From Global Value Chains (GVC) to Innovation Systems for Local Value Chains and Knowledge Creation[J]. European Journal of Development Research,2018,30(3):575.

Keijser C., Iizuka, M. Looking Beyond Global Value Chains in Capacity Development: The Case of the IT-Enabled Service (ITES) Sector in South Africa[J]. European Journal of Development Research,2018,30(3):442-461.

Lema R., Rabellotti, R. & Gehl Sampath, P. Innovation Trajectories in Developing Countries: Co-evolution of Global Value Chains and Innovation Systems[J]. European Journal of Development Research,2018,30(3):345-363.

Upward R., Z. Wang, and J. Zheng, 2013, “Weighing China’s export basket: The domestic content and technology intensity of Chinese exports”, Journal of Comparative Economics, 41(2), 527—543.

WEBB E.Agency costs,leverage,and corporate social responsibility:A test of causality[J].Financial Decisions,2005,3(1):1-19.

Appendix 1

In this study, we employed Principal Component Analysis (PCA) to measure financing environment values. This method reduces dimensionality by extracting a small number of principal components from multiple correlated variables, thereby synthesizing information across variables. The specific analytical steps are as follows:

First, based on the indicator selection by Qi et al. (2024), we identified the following indicators from the raw data:

Table 9. Principal Component Analysis Indicators

Financing capacity	Asset-liability ratio	Total corporate liabilities/Total Assets of the Enterprise
	Total Assets Growth Ratio	Corporate Asset Growth Rate/Total assets of the enterprise for the previous year
Turnover capacity	Current Asset Turnover Ratio	Enterprise's Main Business Revenue/Total Current Assets of the Enterprise
	Total Asset Turnover Ratio	Enterprise Main Business Revenue/Total Assets of the Enterprise
Value added capability	Cost-to-Revenue Ratio	Total corporate profits/Total Enterprise Costs
	Net Profit Margin	(Total corporate profits-Total corporate income tax payable)/Primary Revenue Streams

Author Introduction

Guangyue Ge, Male, born in 2001, obtained a Master's degree by research in International Business from Shanghai University of International Business and Economics in 2025, research interest in International Investment and Transnational Operation, email: geguangyue0814@163.com.

Chen Lin, female, born in 2003, pursuing a master's in international business at the Shanghai University of International Business and Economics, research interest in business big data development and application, email: a15167726601@126.com

Zhuqing Mao, Assistant Professor at Shanghai University of International Business and Economics, Senior Research Associate of SARChI Industrial Development at University of Johannesburg. She obtained her Ph.D. in Economics from Seoul National University. She has subsequently engaged in postdoctoral and visiting research positions at Inha University, University of Johannesburg, and UNU-MERIT. Her research interests include Development Economics, International Trade, Industrial Studies, Area Studies (China, South Korea, Taiwan, and other developing economies), and the Economics of Innovation. Her email is mm7ng7ng@hotmail.com.